

Voice Call Quality Customer Experience Analysis

1. Project Overview

This analysis evaluates user feedback on voice call quality in India, captured via the TRAI MyCall App across 2023. The dataset includes telecom operator ratings, network type, indoor/outdoor usage, and call drop experiences across various Indian states.

2. Data Preparation

Data was aggregated from 10 monthly files, cleaned, and normalized. Entries having the longitude and latitude values as -1 which is not possible and missing the state name were excluded, resulting in 3,458 valid observations.

3. Key Findings

3.1 Best Rated Operators by State

- Vi led in Andhra Pradesh for overall and outdoor ratings.
- Rjio dominated in Bihar, Chhattisgarh, and Chandigarh.
- Airtel was highest-rated in Delhi and also in Chhattisgarh for indoor/outdoor usage.

3.2 Network Type vs. Rating

- 4G networks received the best average ratings.
- 2G and 3G networks lagged significantly behind in performance.

3.3 Call Drop Category vs. Rating

- 'Call Dropped' events had the lowest average rating of 1.21.
- 'Satisfactory' experiences averaged a strong 4.20.

3.4 Indoor vs. Outdoor Call Drops (Hypothesis Test)

A t-test comparing call drop rates showed:

- Indoor call drop rate: 3.52%
- Outdoor call drop rate: 6.54%

4. Dashboard Design (Power BI)

The Power BI dashboard includes:

- State-wise best operator visualization
- Indoor vs. outdoor call analysis
- Network-wise rating distribution
- Call drop analysis with filters by state, operator, and network type.